Separable Equations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Order and verifying solutions**

State the order of the differential equation $y'' + 3y' - 2y = \sin x$.

Second order (the highest derivative appearing is y'').

2 **[Foundational]** • **Target: Order and verifying solutions**

State the order of $\left(\frac{dy}{dx}\right)^3 + xy = 0$, and note that order depends only on the highest derivative, not on any power of it.

First order — the equation involves only y' (cubed), no higher derivative.

3 **[Foundational]** • **Target: Order and verifying solutions**

Verify that $y = e^{2x}$ is a solution of $y' - 2y = 0$.

$y' = 2e^{2x}$, so $y' - 2y = 2e^{2x} - 2e^{2x} = 0$. Yes, it is a solution.

4 **[Foundational]** • **Target: Equilibrium solutions and qualitative behavior**

Find the equilibrium (constant) solutions of $\frac{dy}{dx} = y(3 - y)$.

Set $y(3 - y) = 0$: the equilibrium solutions are $y = 0$ and $y = 3$.

5 **[Foundational]** • **Target: Equilibrium solutions and qualitative behavior**

For $\frac{dy}{dx} = y(3 - y)$, determine whether $y = 0$ and $y = 3$ are stable or unstable by checking the sign of $\frac{dy}{dx}$ just above and below each.

For $0 < y < 3$: $y(3 - y) > 0$, so solutions increase away from $y = 0$ and toward $y = 3$. For $y > 3$: $y(3 - y) < 0$, so solutions decrease back toward $y = 3$. Hence $y = 0$ is unstable and $y = 3$ is stable.

6 **[Foundational]** • **Target: Equilibrium solutions and qualitative behavior**

Find the equilibrium solutions of $\frac{dy}{dx} = (y - 2)(y + 1)$ and classify each as stable or unstable.

Equilibria: $y = 2$ and $y = -1$. For $-1 < y < 2$, $(y - 2)(y + 1) < 0$, so solutions decrease toward $y = -1$: $y = -1$ is stable. For $y > 2$, the product is positive, pushing solutions away: $y = 2$ is unstable.

7 **[Foundational]** • **Target: General versus particular solutions**

Find the general solution of $\frac{dy}{dx} = 2x$, then find the particular solution satisfying $y(1) = 3$.

General solution: $y = x^2 + C$ (this one is already separated, with $\frac{dy}{dx}$ depending only on x). Since $y(1) = 3$: $1 + C = 3 \Rightarrow C = 2$. Particular solution: $y = x^2 + 2$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: General versus particular solutions**

Explain, in terms of the arbitrary constant C, why a differential equation typically has infinitely many solutions but an initial value problem typically has exactly one.

Integrating introduces one arbitrary constant C, so the general solution is really a one-parameter family of curves, all satisfying the ODE. An initial condition $y(x_0) = y_0$ picks out the single curve in that family passing through (x_0, y_0), which (under standard existence/uniqueness conditions) fixes C to exactly one value.

9 **[Intermediate]** • **Target: Application: radioactive decay and population growth**

A radioactive substance decays according to $\frac{dA}{dt} = -kA$. If $A(0) = 100$ g and $A(5) = 50$ g, find k.

$A = 100e^{-kt}$. Then $50 = 100e^{-5k} \Rightarrow e^{-5k} = \frac{1}{2} \Rightarrow -5k = \ln\frac{1}{2} \Rightarrow k = \frac{\ln 2}{5} \approx 0.1386$ per unit time.

10 **[Intermediate]** • **Target: Application: radioactive decay and population growth**

A population grows according to $\frac{dP}{dt} = 0.03P$, with $P(0) = 200$. Find $P(10)$.

$P(t) = 200e^{0.03t}$, so $P(10) = 200e^{0.3} \approx 269.97$.

11 **[Intermediate]** • **Target: Application: radioactive decay and population growth**

A bank account earns continuously-compounded interest, $\frac{dA}{dt} = 0.05A$, with $A(0) = 1000$. Find $A(20)$ and the doubling time.

$A(t) = 1000e^{0.05t}$, so $A(20) = 1000e^{1} \approx 2718.28$. Doubling time solves $e^{0.05t} = 2 \Rightarrow t = \frac{\ln 2}{0.05} \approx 13.86$ years.

12 **[Intermediate]** • **Target: Application: Newton's Law of Cooling**

Newton's Law of Cooling gives $\frac{dT}{dt} = -k(T - T_s)$, where T_s is the surroundings' temperature. Separate variables and integrate to find the general solution.

Separate: $\frac{dT}{T - T_s} = -k\,dt$. Integrate: $\ln|T - T_s| = -kt + C$. So $T = T_s + Ae^{-kt}$.

13 **[Intermediate]** • **Target: Application: Newton's Law of Cooling**

A cup of coffee at $90\,^\circ$C is left in a $20\,^\circ$C room. If $T(10) = 60\,^\circ$C, find k, then find $T(20)$.

$T = 20 + Ae^{-kt}$; $T(0) = 90 \Rightarrow A = 70$. $T(10) = 60 \Rightarrow 40 = 70e^{-10k} \Rightarrow k = \frac{1}{10}\ln\frac{7}{4} \approx 0.056$. Then $T(20) = 20 + 70e^{-20k} = 20 + 70\left(\frac{4}{7}\right)^2 = 20 + 70 \cdot \frac{16}{49} \approx 42.86\,^\circ$C.

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14 [Intermediate] • Target: Mixed review

Determine whether $\frac{dy}{dx} = x + y$ is separable. Justify your answer.

No. The right side $x + y$ cannot be factored as (a function of x) $\times$ (a function of y), since it is a sum, not a product.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Mixed review • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

A tank of water drains according to Torricelli's Law, $\frac{dh}{dt} = -kh^{1/2}$, where h is the water height. Separate variables and integrate.

Separate: $h^{-1/2}\,dh = -k\,dt$. Integrate: $2h^{1/2} = -kt + C$, so $h = \left(\frac{C - kt}{2}\right)^2$.

16 [Challenging] • Target: Mixed review • Cross-Lesson: 5.01 Variation of Parameters

For the model $\frac{dy}{dx} = \frac{2x}{y}$ with $y(0) = 3$, find the particular solution and state its domain.

Separate: $y\,dy = 2x\,dx \Rightarrow \frac{y^2}{2} = x^2 + C$. Since $y(0) = 3$: $\frac{9}{2} = C$, so $y^2 = 4x^2 + 9$, and since $y(0) = 3 > 0$: $y = \sqrt{4x^2 + 9}$, defined for all real x.

17 [Challenging] • Target: Further practice: separable equations • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Find all equilibrium solutions to $\frac{dy}{dx} = y(4 - y) - 3$, classify their stability, and find the particular solution passing through $y(0) = 2$.

Step 1: Locate equilibria: $-y^2 + 4y - 3 = 0 \Rightarrow (y - 1)(3 - y) = 0 \Rightarrow y_1 = 1, \quad y_2 = 3.$ Step 2: Stability via derivative: Let $f(y) = -y^2 + 4y - 3$. Then $f'(y) = -2y + 4$. At $y = 1$, $f'(1) = 2 > 0 \Rightarrow y = 1$ is an unstable source. At $y = 3$, x: $\frac{1}{(y-1)(3-y)}\,dy = dx \Rightarrow \frac{1}{2}\left(\frac{1}{y-1} + \frac{1}{3-y}\right)dy = dx.$ Integrating.

18 [Challenging] • Target: Mixed review • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Compute the first three Picard successive approximations $y_1(x), y_2(x), y_3(x)$ for the initial value problem $\frac{dy}{dx} = 2x(1 + y)$ with $y(0) = 0$, and identify the Taylor series of the exact solution.

Picard Integral Operator: $y_{k+1}(x) = y_0 + \int_0^x 2t(1 + y_k(t))\,dt, \qquad y_0(x) = 0.$ Iterate 1: $y_1(x) = 0 + \int_0^x 2t(1 + 0)\,dt = [t^2]_0^x = x^2.$ Iterate 2: $y_2(x) = \int_0^x 2t(1 + t^2)\,dt = \int_0^x (2t + 2t^3)\,dt = x^2 + \frac{x^4}{2}.$ Iterate 3: $y_3(x) = \int_0^x 2t\left(1 + t^2 + \frac{t^4}{2}\right)dt = \int_0^x (2t + 2t^3 + t^5)\,dt = x^2 + \frac{x^4}{2} + \frac{x^6}{6}.$ Exact Solution: By inspection, $y_k(x) = \sum_{m=1}^{k} \frac{(x^2)^m}{m!} \to e^{x^2} - 1.$ Verify.

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19 [Challenging] • Target: Order and verifying solutions • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

Solve the initial value problem $\frac{dy}{dx} = 2x(y^2 + 1)$ with $y(0) = 1$, and determine its maximal interval of existence.

Step 1: Separate variables: $\frac{1}{y^2+1}\,dy = 2x\,dx$. Step 2: Integrate both sides: $\int \frac{1}{y^2+1}\,dy = \int 2x\,dx \Rightarrow \arctan(y) = x^2 + C$. Step 3: Apply the initial condition $y(0) = 1$: $\arctan(1) = 0^2 + C \Rightarrow C = \frac{\pi}{4}$. Step 4: Solve explicitly for $y(x)$: $y(x) = \tan\left(x^2 + \frac{\pi}{4}\right)$. Step 5: Interval of existence: The tangent function has vertical asymptotes where its argument equals $\pm\frac{\pi}{2}$. To contain $x_0 = 0$: \[-\frac{\pi}{2}

20 [Challenging] • Target: Recognizing and solving separable equations • Cross-Lesson: 5.01 Variation of Parameters

[First-Order Differential Equations, Separable Equations & Existence Theory · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Order and verifying solutions

A hot object cools from 100°C to 80°C in 10 minutes in an environment kept at 20°C. Under Newton's Law of Cooling, at what time t is the instantaneous rate of cooling $|dT/dt|$ exactly half of its initial value at $t = 0$?

a. $10\frac{\ln 2}{\ln(4/3)}$ ✓

b. $10\frac{\ln(4/3)}{\ln 2}$

c. 20 minutes

d. $5\frac{\ln 2}{\ln(4/3)}$

Separable: $\frac{dT}{dt} = -k(T - 20) \Rightarrow |T'(t)| = k(T(t) - 20) = 80ke^{-kt}$. The initial cooling rate is $80k$. The rate is halved when $e^{-kt} = 1/2 \Rightarrow t = \frac{\ln 2}{k}$. From $T(10) - 20 = 60 = 80e^{-10k}$, we find $e^{-10k} = \frac{3}{4} \Rightarrow k = \frac{1}{10}\ln(4/3)$. Thus $t = 10\frac{\ln 2}{\ln(4/3)}$.

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22 [Foundational] • Target: Recognizing and solving separable equations

An object cools in a room kept at a constant temperature T_a. At $t = 0$, its temperature is $100\,^\circ\text{C}$; at $t = 10$ min, it is $60\,^\circ\text{C}$; and at $t = 20$ min, it is $40\,^\circ\text{C}$. Assuming Newton's Law of Cooling, what is the room temperature T_a?

a. $20\,^\circ\text{C}$ ✓

b. $15\,^\circ\text{C}$

c. $25\,^\circ\text{C}$

d. $18\,^\circ\text{C}$

Separable: $\frac{dT}{dt} = -k(T - T_a) \Rightarrow T(t) - T_a = (100 - T_a)e^{-kt}$. At $t = 10$, $60 - T_a = (100 - T_a)e^{-10k}$. At $t = 20$, $40 - T_a = (100 - T_a)e^{-20k} = (100 - T_a)(e^{-10k})^2$. Thus $\frac{40 - T_a}{60 - T_a} = e^{-10k} = \frac{60 - T_a}{100 - T_a}$. Cross-multiplying gives $(60 - T_a)^2 = (40 - T_a)(100 - T_a) \Rightarrow 3600 - 120T_a + T_a^2 = 4000 - 140T_a + T_a^2 \Rightarrow 20T_a = 400 \Rightarrow T_a = 20\,^\circ\text{C}$.

23 [Foundational] • Target: Equilibrium solutions and qualitative behavior

Hot tea cools from $95\,^\circ\text{C}$ to $80\,^\circ\text{C}$ in 5 minutes in a room kept at $20\,^\circ\text{C}$. Under Newton's Law of Cooling, how many additional minutes will it take to cool from $80\,^\circ\text{C}$ to $50\,^\circ\text{C}$?

a. $10\frac{\ln 2}{\ln(5/4)}$

b. $5\frac{\ln(4/5)}{\ln 2}$

c. $5\frac{\ln 2}{\ln(5/4)}$ ✓

d. $5\frac{\ln(5/2)}{\ln(5/4)}$

Separable: $T(t) - 20 = 75e^{-kt}$. At $t = 5$, $80 - 20 = 60 = 75e^{-5k} \Rightarrow e^{-5k} = \frac{4}{5} \Rightarrow k = \frac{1}{5}\ln(5/4)$. For $T(t) = 50$, $30 = 75e^{-kt} \Rightarrow e^{-kt} = \frac{2}{5}$, so $t = \frac{\ln(5/2)}{k} = 5\frac{\ln(5/2)}{\ln(5/4)}$. The additional time is $t - 5 = 5\left(\frac{\ln(5/2)}{\ln(5/4)} - 1\right) = 5\frac{\ln(5/2) - \ln(5/4)}{\ln(5/4)} = 5\frac{\ln 2}{\ln(5/4)}$.

24 [Foundational] • Target: General versus particular solutions

A metal ingot at $800\,^\circ\text{C}$ is immersed in a cooling bath kept at $50\,^\circ\text{C}$. After 1 minute, its temperature drops to $500\,^\circ\text{C}$. Under Newton's Law of Cooling, how many additional minutes will it take to cool to $100\,^\circ\text{C}$?

a. $\frac{3\ln 2}{\ln(5/3)}$

b. $\frac{\ln 3}{\ln(5/3)}$

c. $\frac{2\ln 3}{\ln(5/3)}$ ✓

d. $\frac{\ln 15}{\ln(5/3)}$

Separable: $T(t) - 50 = 750e^{-kt}$. At $t = 1$, $450 = 750e^{-k} \Rightarrow e^{-k} = \frac{3}{5} \Rightarrow k = \ln(5/3)$. For $T(t) = 100$, $50 = 750e^{-kt} \Rightarrow e^{-kt} = \frac{1}{15} \Rightarrow t = \frac{\ln 15}{\ln(5/3)}$. The additional time required is $t - 1 = \frac{\ln 15 - \ln(5/3)}{\ln(5/3)} = \frac{\ln(15 \cdot 3/5)}{\ln(5/3)} = \frac{\ln 9}{\ln(5/3)} = \frac{2\ln 3}{\ln(5/3)}$.

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25 [Foundational] • Target: Initial value problems

An object cools in an environment whose ambient temperature rises linearly according to $T_m(t) = 10 + 2t$ (in $^\circ$C). Newton's Law of Cooling gives $T' = -k(T - T_m(t))$ with $k = 1$ hr^{-1}. If $T(0) = 50\,^\circ$C, what is $T(1)$?

a. $10 + 42e^{-1}$ ✓

b. $10 + 40e^{-1}$

c. $12 + 42e^{-1}$

d. $8 + 42e^{-1}$

Linear ODE: $T' + T = 10 + 2t$. Integrating factor: $\mu = e^t$. Multiplying gives $(Te^t)' = (2t + 10)e^t$. Integrating by parts yields $Te^t = (2t + 8)e^t + C \Rightarrow T(t) = 2t + 8 + Ce^{-t}$. Using $T(0) = 50$ yields $50 = 8 + C \Rightarrow C = 42$. At $t = 1$, $T(1) = 2(1) + 8 + 42e^{-1} = 10 + 42e^{-1}$.

26 [Foundational] • Target: Application: radioactive decay and population growth

A cup of coffee at $90\,^\circ$C is placed in a room held at $20\,^\circ$C. After 10 minutes, the temperature of the coffee is $60\,^\circ$C. Assuming Newton's Law of Cooling, at what time t (in minutes) will its temperature reach $30\,^\circ$C?

a. $10\frac{\ln 7}{\ln(4/7)}$

b. $20\frac{\ln 7}{\ln(7/4)}$

c. $10\frac{\ln 7}{\ln(7/4)}$ ✓

d. $10\frac{\ln 4}{\ln(7/4)}$

Separable: $\frac{dT}{dt} = -k(T - 20) \Rightarrow T(t) = 20 + 70e^{-kt}$. At $t = 10$, $60 = 20 + 70e^{-10k} \Rightarrow e^{-10k} = \frac{4}{7}$. For $T(t) = 30$, $30 = 20 + 70e^{-kt} \Rightarrow e^{-kt} = \frac{1}{7}$. Taking logs: $-kt = -\frac{t}{10}\ln(7/4) = -\ln 7 \Rightarrow t = 10\frac{\ln 7}{\ln(7/4)}$.

27 [Foundational] • Target: Application: Newton's Law of Cooling

A thermometer initially reading $25\,^\circ$C is taken outside into air at constant temperature T_m. After 1 minute it reads $15\,^\circ$C, and after 2 minutes it reads $9\,^\circ$C. Assuming Newton's Law of Cooling, what is the outdoor temperature T_m?

a. $-5\,^\circ$C

b. $5\,^\circ$C

c. $3\,^\circ$C

d. $0\,^\circ$C ✓

Separable: $T(t) - T_m = (25 - T_m)e^{-kt}$. At $t = 1$, $15 - T_m = (25 - T_m)e^{-k}$. At $t = 2$, $9 - T_m = (25 - T_m)e^{-2k}$. Squaring the first ratio gives $(15 - T_m)^2 = (25 - T_m)(9 - T_m) \Rightarrow 225 - 30T_m + T_m^2 = 225 - 34T_m + T_m^2 \Rightarrow 4T_m = 0 \Rightarrow T_m = 0\,^\circ$C.

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28 [Foundational] • Target: Further practice: separable equations

Under Newton's Law of Cooling, $\frac{dT}{dt} = -k(T - T_m)$ with surrounding temperature $T_m = 20\,^\circ\text{C}$. An object starts at $T(0) = 80\,^\circ\text{C}$ and cools to $T(1) = 50\,^\circ\text{C}$ at $t = 1$ hour. What is the cooling constant k?

a. $-\ln 2$

b. $\ln 2$ ✓

c. $2\ln 2$

d. $\ln(5/8)$

Separable: $\frac{dT}{T-20} = -k\,dt \Rightarrow T(t) - 20 = (80-20)e^{-kt} = 60e^{-kt}$. At $t = 1$, $50 - 20 = 60e^{-k} \Rightarrow 30 = 60e^{-k} \Rightarrow e^{-k} = \frac{1}{2} \Rightarrow k = \ln 2$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Mixed review

Consider the differential equation $\frac{dy}{dx} = \frac{x^2\cos(y)}{1-x^3}$ with the initial condition $y(0) = \frac{\pi}{2}$. If there exists a unique solution in a neighborhood of $x = 0$ and $y = \frac{\pi}{2}$, find the value of $y'(0)$.

a. **0** ✓

b. 1

c. $\frac{\pi}{2}$

d. Undefined

The given differential equation is $\frac{dy}{dx} = \frac{x^2\cos(y)}{1-x^3}$ with the initial condition $y(0) = \frac{\pi}{2}$. We are asked to find $y'(0)$. The differential equation directly gives us the expression for $\frac{dy}{dx}$, which is $y'(x)$. To find $y'(0)$, we substitute $x = 0$ and $y = y(0) = \frac{\pi}{2}$ into the differential equation: $y'(0) = \frac{(0)^2\cos(y(0))}{1-(0)^3}$ Substitute $y(0) = \frac{\pi}{2}$: $y'(0) = \frac{0 \cdot \cos(\frac{\pi}{2})}{1-0}$ We know that.

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30 [Intermediate] • Target: Order and verifying solutions

Consider the nonhomogeneous second-order linear differential equation $y'' - 5y' + 6y = 12x + 8$. $y_h(x) = c_1e^{2x} + c_2e^{3x}$ is the general solution to the corresponding homogeneous equation. Find the particular solution $y_p(x)$ using the method of undetermined coefficients, and then determine the value of the constant A such that $y_p(x) = Ax + B$ for some constant B.

a. $A = 2$ ✓

b. $A = 3$

c. $A = 6$

d. $A = 12$

The given nonhomogeneous differential equation is $y'' - 5y' + 6y = 12x + 8$. The form of the nonhomogeneous term is $g(x) = 12x + 8$, which is a polynomial of degree 1. Thus, we assume a particular solution of the form $y_p(x) = Ax + B$. We need to find the first and second derivatives of $y_p(x)$: $y_p'(x) = A$ $y_p''(x) = 0$ Substitute these into the differential equation: $0 - 5(A) + 6(Ax + B) = 12x + 8$ $-5A + 6Ax + 6B = 12x + 8$ $6Ax + (6B - 5A) = 12x + 8$ Equating the coefficients of like powers of x: For.

31 [Intermediate] • Target: Recognizing and solving separable equations

Consider the second-order nonhomogeneous linear differential equation $y'' - 5y' + 6y = 12x + 4$. If a particular solution is of the form $y_p(x) = Ax + B$ and the general solution is $y(x) = C_1e^{2x} + C_2e^{3x} + Ax + B$, find the value of A.

a. 1

b. **2** ✓

c. 6

d. 12

The given differential equation is $y'' - 5y' + 6y = 12x + 4$. \\ We are given that a particular solution is of the form $y_p(x) = Ax + B$. \\ To find the coefficients A and B, we substitute $y_p(x)$ and its derivatives into the differential equation. \\ The first derivative is $y_p'(x) = A$. \\ The second derivative is $y_p''(x) = 0$. \\ Substituting these into the differential equation: \\ xBy comparing the coefficients of x on both sides, we get: \\ $6A = 12 \Rightarrow A = 2$By comparing the constant te.

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32 [Intermediate] • Target: Equilibrium solutions and qualitative behavior

Consider the separable differential equation $\frac{dy}{dx} = 2x$ with the initial condition $y(0) = 3$. Find the value of $y(2)$.

a. **7** ✓

b. 6

c. 8

d. 5

The differential equation is $\frac{dy}{dx} = 2x$. This is a separable equation. We can rewrite it as $dy = 2x\,dx$. Integrating both sides gives $\int dy = \int 2x\,dx$, which results in $y = x^2 + C$, where C is the constant of integration. We are given the initial condition $y(0) = 3$. Substituting these values into the general solution, we get $3 = (0)^2 + C$, so $C = 3$. Therefore, the particular solution is $y = x^2 + 3$. We need to find the value of $y(2)$. Substituting $x = 2$ into the particular soluti.

33 [Intermediate] • Target: General versus particular solutions

A population grows according to $\frac{dP}{dt} = \frac{t}{1+t^2}P$ with initial size $P(0) = 500$. At what time $t > 0$ does the population reach 1300?

a. 2.6

b. $\sqrt{5}$

c. 3.2

d. 2.4 ✓

Separable: $\frac{dP}{P} = \frac{t}{1+t^2}\,dt$. Integrating gives $\ln(P/500) = \frac{1}{2}\ln(1+t^2) \Rightarrow P(t) = 500\sqrt{1+t^2}$. Setting $P(t) = 1300$ gives $500\sqrt{1+t^2} = 1300 \Rightarrow \sqrt{1+t^2} = \frac{13}{5} \Rightarrow 1+t^2 = \frac{169}{25} \Rightarrow t^2 = \frac{144}{25} \Rightarrow t = \frac{12}{5} = 2.4$.

34 [Intermediate] • Target: Initial value problems

A population model exhibits accelerated growth satisfying $\frac{dP}{dt} = 0.02P^2$ with $P(0) = 50$. At what time t does the population experience finite-time blow-up (i.e., $P(t) \to \infty$)?

a. $t = 50$

b. $t = 1$ ✓

c. $t = 0.5$

d. $t = 2$

Separable: $P^{-2}\,dP = 0.02\,dt$. Integrating gives $-\frac{1}{P} = 0.02t + C$. Using $P(0) = 50$ yields $-\frac{1}{50} = C$. Thus $\frac{1}{P} = \frac{1}{50} - 0.02t = \frac{1-t}{50} \Rightarrow P(t) = \frac{50}{1-t}$. Blow-up occurs when the denominator is zero: $1 - t = 0 \Rightarrow t = 1$.

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35 **[Intermediate]** • **Target: Application: radioactive decay and population growth**

A population of bacteria grows at a rate proportional to its size: $\frac{dP}{dt} = kP$. If the population triples every 4 hours, how many hours does it take to grow to 9 times its initial size?

a. 6

b. 16

c. 12

d. 8 ✓

Separable: $P(t) = P_0 e^{kt}$. Tripling every 4 hours means $P(4) = 3P_0 \Rightarrow e^{4k} = 3$. To reach $9P_0 = 3^2 P_0$, we set $e^{kt} = 3^2 = (e^{4k})^2 = e^{8k}$. Thus $t = 8$ hours.

36 **[Intermediate]** • **Target: Application: Newton's Law of Cooling**

What is the general solution to the initial value problem $\frac{dy}{dt} = -3y$ with $y(0) = 50$?

a. $y(t) = 50 - e^{-3t}$

b. $y(t) = 50e^{3t}$

c. $y(t) = 50e^{-3t}$ ✓

d. $y(t) = -3t + 50$

Separable: $\frac{dy}{y} = -3\,dt \Rightarrow \ln|y| = -3t + C \Rightarrow y(t) = y(0)e^{-3t}$. Using $y(0) = 50$ yields $y(t) = 50e^{-3t}$.

37 **[Intermediate]** • **Target: Further practice: separable equations**

Consider the differential equation $\frac{dy}{dx} = kx(y^2 - 1)$, where k is a positive constant. If the solution passing through $(1, 2)$ also passes through $(2, 3)$, determine the value of the constant k.

a. $\frac{1}{4}\ln\left(\frac{3}{2}\right)$

b. $\frac{1}{3}\ln\left(\frac{2}{3}\right)$

c. $\frac{1}{4}\ln\left(\frac{2}{3}\right)$ ✓

d. $\frac{1}{3}\ln\left(\frac{3}{2}\right)$

The given differential equation is separable: $\frac{dy}{y^2-1} = kx\,dx$. Integrating both sides: $\int \frac{dy}{y^2-1} = \int kx\,dx$ Using partial fractions for the left side, $\frac{1}{y^2-1} = \frac{1}{2}\left(\frac{1}{y-1} - \frac{1}{y+1}\right)$. So, $\frac{1}{2}\int\left(\frac{1}{y-1} - \frac{1}{y+1}\right)dy = \frac{1}{2}\ln\left|\frac{y-1}{y+1}\right|$. The right side integrates to $\frac{1}{2}kx^2 + C_1$. Thus, $\frac{1}{2}\ln\left|\frac{y-1}{y+1}\right| = \frac{1}{2}kx^2 + C_1$, w.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Mixed review • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Consider the second-order linear homogeneous differential equation $y'' - 2y' + y = 0$. \\ If one of the fundamental solutions is $y_1(x) = e^x$, and the Wronskian of two linearly independent solutions $y_1(x)$ and $y_2(x)$ is given by $W(y_1, y_2)(x) = 5e^{2x}$, find the unknown parameter k in $y_2(x) = kxe^x$.

a. $k = 5$ ✓

b. $k = -5$

c. $k = 1$

d. $k = -1$

The given differential equation is $y'' - 2y' + y = 0$. The characteristic equation is $r^2 - 2r + 1 = 0$, which factors as $(r-1)^2 = 0$. Thus, the roots are repeated: $r_1 = r_2 = 1$. \\ One fundamental solution is given as $y_1(x) = e^x$. For a second-order linear homogeneous equation $y'' + p(x)y' + q(x)y = 0$ with a known solution $y_1(x)$, the Wronskian is given by $W(y_1, y_2)(x) = y_1(x)y_2'(x) - y_1'(x)y_2(x) = Ce^{-\int p(x)dx}$. \\ In our equation, $p(x) = -2$. So, $-\int p(x)dx = -\int(-2)dx = 2x$..

39 [Challenging] • Target: Order and verifying solutions • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

Consider the separable differential equation $\frac{dy}{dx} = ky$ where k is a positive constant. A solution curve passes through the points $(0, 5)$ and $(2, 20)$. Determine the value of k and then find the value of y when $x = 3$.

a. $k = \ln 2, y(3) = 40$

b. $k = 2\ln 2, y(3) = 40$

c. $k = \ln 2, y(3) = 20\sqrt{2}$

d. $k = \ln 4, y(3) = 80$ ✓

The differential equation $\frac{dy}{dx} = ky$ is separable. Separating variables gives $\frac{dy}{y} = k\,dx$. Integrating both sides, we get $\int \frac{dy}{y} = \int k\,dx$, which yields $\ln|y| = kx + C_1$. Exponentiating both sides gives $|y| = e^{kx + C_1} = e^{C_1}e^{kx}$. Let $C = \pm e^{C_1}$, so $y = Ce^{kx}$. We are given that the solution curve passes through $(0, 5)$. Substituting these values into $y = Ce^{kx}$: $5 = Ce^{k(0)} \Rightarrow 5 = C$. Thus, the particular solution is $y = 5e^{kx}$. We ar.

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40 [Challenging] • Target: Recognizing and solving separable equations • Cross-Lesson: 5.01 Variation of Parameters

Consider the separable first-order differential equation $\frac{dy}{dx} = \frac{x^2}{y}$. If the solution passes through the point $(1, 2)$, find the explicit solution for $y(x)$ assuming $y > 0$.

a. $y = \sqrt{\frac{x^3}{3} + \frac{7}{3}}$ ✓

b. $y = \sqrt{\frac{x^3}{3} + \frac{5}{3}}$

c. $y = \sqrt{\frac{x^3}{3} + 2}$

d. $y = \sqrt{\frac{x^3}{3} + \frac{2}{3}}$

The given differential equation is $\frac{dy}{dx} = \frac{x^2}{y}$. This is a separable equation. We can rewrite it as $y\,dy = x^2\,dx$. Integrating both sides, we get: $\int y\,dy = \int x^2\,dx$ $\frac{1}{2}y^2 = \frac{1}{3}x^3 + C$ We are given that the solution passes through the point $(1, 2)$. Substituting these values into the integrated equation: $\frac{1}{2}(2)^2 = \frac{1}{3}(1)^3 + C$ $\frac{1}{2}(4) = \frac{1}{3} + C$ $2 = \frac{1}{3} + C$ Solving for C.

41 [Challenging] • Target: Equilibrium solutions and qualitative behavior • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Consider the second-order linear non-homogeneous differential equation: $\frac{d^2y}{dx^2} + 4y = \sec(2x)$. Suppose a particular solution is found using the method of variation of parameters. Let the fundamental set of solutions for the associated homogeneous equation be $y_1(x) = \cos(2x)$ and $y_2(x) = \sin(2x)$. If the particular solution is expressed in the form $y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$, determine the value of $u_2'(x)$.

a. $\frac{1}{2}$ ✓

b. $\frac{1}{4}$

c. $\frac{1}{2}\sin(2x)$

d. $\frac{1}{4}\cos(2x)$

The given differential equation is $\frac{d^2y}{dx^2} + 4y = \sec(2x)$. The associated homogeneous equation is $\frac{d^2y}{dx^2} + 4y = 0$. The characteristic equation is $r^2 + 4 = 0$, which has roots $r = \pm 2i$. Thus, the fundamental set of solutions for the homogeneous equation is $y_1(x) = \cos(2x)$ and $y_2(x) = \sin(2x)$. We use the method of variation of parameters to find a particular solution of the form $y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$. The Wronskian of y_1 and y_2 is.

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42 [Challenging] • Target: General versus particular solutions • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Consider the second-order nonhomogeneous linear differential equation $y'' - 2y' + y = e^x \sin(x)$. Let $y_p(x)$ be a particular solution obtained using the method of undetermined coefficients. Determine the value of $y_p''(\pi) - 2y_p'(\pi) + y_p(\pi)$.

a. 0 ✓

b. e^{π}

c. $-\pi e^{\pi}$

d. $e^{\pi}\sin(\pi)$

The problem asks for the value of the left-hand side of the differential equation $y'' - 2y' + y = e^x \sin(x)$ when evaluated at a particular solution $y_p(x)$. By definition, a particular solution $y_p(x)$ must satisfy the nonhomogeneous differential equation for all x. Therefore, when we substitute $y_p(x)$ into the differential equation, the left-hand side must equal the right-hand side. Specifically, for any particular solution $y_p(x)$, it must be true that $y_p''(x) - 2y_p'(x) + y_p(x) = e^x \sin(x)$ for a.

43 [Challenging] • Target: Initial value problems • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

Consider a second-order linear non-homogeneous differential equation of the form $y''(x) + p(x)y'(x) + q(x)y(x) = g(x)$. If $y_1(x)$ and $y_2(x)$ form a fundamental set of solutions to the corresponding homogeneous equation $y''(x) + p(x)y'(x) + q(x)y(x) = 0$, and the Wronskian is $W(y_1, y_2)(x) = y_1(x)y_2'(x) - y_1'(x)y_2(x)$, which of the following expressions represents a particular solution $y_p(x)$ obtained using the method of variation of parameters?

a. $y_p(x) = -y_1(x)\int \frac{y_2(x)g(x)}{W(y_1,y_2)(x)}\,dx + y_2(x)\int \frac{y_1(x)g(x)}{W(y_1,y_2)(x)}\,dx$ ✓

b. $y_p(x) = y_1(x)\int \frac{y_2(x)g(x)}{W(y_1,y_2)(x)}\,dx + y_2(x)\int \frac{y_1(x)g(x)}{W(y_1,y_2)(x)}\,dx$

c. $y_p(x) = -y_1(x)\int \frac{y_2(x)g(x)}{W(y_1,y_2)(x)}\,dx - y_2(x)\int \frac{y_1(x)g(x)}{W(y_1,y_2)(x)}\,dx$

d. $y_p(x) = y_1(x)\int \frac{y_2(x)g(x)}{W(y_1,y_2)(x)}\,dx - y_2(x)\int \frac{y_1(x)g(x)}{W(y_1,y_2)(x)}\,dx$

The method of variation of parameters for a second-order linear non-homogeneous differential equation $y''(x) + p(x)y'(x) + q(x)y(x) = g(x)$, where $y_1(x)$ and $y_2(x)$ are linearly independent solutions to the homogeneous equation, seeks a particular solution of the form $y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$. The standard formulas for $u_1'(x)$ and $u_2'(x)$ are derived as follows: $u_1'(x) = -\frac{y_2(x)g(x)}{W(y_1,y_2)(x)}$ $u_2'(x) = \frac{y_1(x)g(x)}{W(y_1,y_2)(x)}$ Integrating these expressions gives.

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44 [Challenging] • Target: Application: radioactive decay and population growth • Cross-Lesson: 5.01 Variation of Parameters

Consider the second-order linear homogeneous differential equation: $y''(x) - ky'(x) + 4y(x) = 0$. If the characteristic equation has a repeated real root, and one of the solutions to the differential equation is $y_1(x) = xe^{2x}$, determine the value of the parameter k.

a. 2

b. **4** ✓

c. 3

d. 1

The given differential equation is $y''(x) - ky'(x) + 4y(x) = 0$. The auxiliary equation is $r^2 - kr + 4 = 0$. For the auxiliary equation to have a repeated real root, the discriminant must be zero: $D = (-k)^2 - 4(1)(4) = 0$. This implies $k^2 - 16 = 0$, so $k = \pm 4$. If $k = 4$, the auxiliary equation is $r^2 - 4r + 4 = 0$, which factors as $(r - 2)^2 = 0$, yielding a repeated real root $r = 2$. The general solution for a repeated real root r is $y(x) = c_1e^{rx} + c_2xe^{rx}$. In this case.

45 [Challenging] • Target: Application: Newton's Law of Cooling • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Consider the linear first-order differential equation $\frac{dy}{dx} + 2y = 6$. If the initial condition is $y(0) = 1$, find the value of $y(1)$.

a. 2

b. $e^2 - 1$

c. $3 - 2e^{-2}$

d. $3 - 2e^{-1}$ ✓

The given differential equation is a linear first-order ODE of the form $\frac{dy}{dx} + P(x)y = Q(x)$, where $P(x) = 2$ and $Q(x) = 6$. The integrating factor $\mu(x)$ is given by $\mu(x) = e^{\int P(x)dx}$. Calculating the integrating factor: $\mu(x) = e^{\int 2dx} = e^{2x}$. Multiplying the differential equation by the integrating factor: $e^{2x}\frac{dy}{dx} + 2e^{2x}y = 6e^{2x}$. The left side is the derivative of the product $y\mu(x)$: $\frac{d}{dx}(ye^{2x}) = 6e^{2x}$. Integrating both sides with respect t.

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